

Lesson Plan: Mathematics

Period 1: Introduction to Linear Equations and Properties of Equality (40 min)

Objectives: Define a linear equation in one variable.; Identify the variable, constant, and coefficient in a given linear equation.; Explain the properties of equality (addition, subtraction, multiplication, division) using the balance analogy.; Distinguish between an expression and an equation.

Concepts: Linear Equation in One Variable, Properties of Equality

Period 2: Solving Simple Linear Equations (40 min)

Objectives: Apply the properties of equality to solve one-step and two-step linear equations.; Demonstrate the balance method to isolate the variable.; Verify the obtained solution by substituting it back into the original equation.; Solve simple linear equations involving integers.

Concepts: Properties of Equality, Solution of an Equation

Period 3: The Transposition Method (40 min)

Objectives: Explain transposition as a shortcut derived from the properties of equality.; Apply transposition to move terms across the equality sign, changing their sign appropriately.; Solve linear equations efficiently using transposition.; Verify solutions obtained through transposition.

Concepts: Transposition, Solution of an Equation

Period 4: Equations with Variables on Both Sides (45 min)

Objectives: Describe the structure of an equation having variables on both sides.; Apply transposition to collect variable terms on one side and constants on the other.; Solve equations with variables on both sides accurately.; Analyze solutions to check for errors by verifying both sides of the equation are equal.

Concepts: Equations with Variables on Both Sides, Transposition, Solution of an Equation

Period 5: Applying Linear Equations to Word Problems (45 min)

Objectives: Translate real-life word problems into linear equations in one variable.; Apply transposition and

properties of equality to solve the resulting equations.; Evaluate the reasonableness of the solution in the context of the original problem.; Create and solve original word problems involving linear equations, including those with variables on both sides.

Concepts: Linear Equation in One Variable, Equations with Variables on Both Sides, Solution of an Equation